Do Now · retrieval

Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Simplify ¬(A ∨ B).

Q2

Keys 7 and 12 hash to index 2 in a table. What is that called?

Q3

What is 1 XOR 1?

Q4

If A changes from 0 to 1 while B is always 1, which variable is constant?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Read the labels — Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

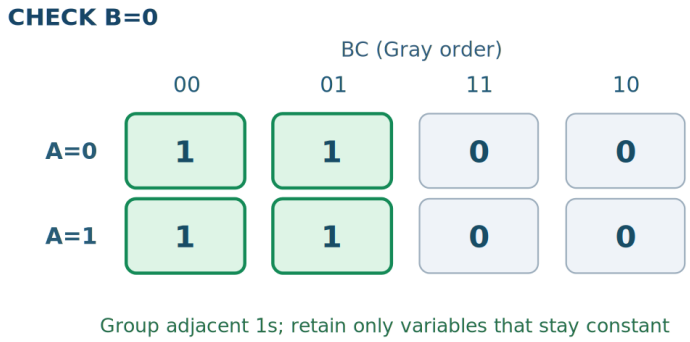
Core idea 2

Group neighbouring ones — The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

Core idea 3

Keep what is constant — A and C change through the group, but B remains 0. So the simplified result is ¬B.

Key visual · complete concept



Group adjacent 1s; retain only variables that stay constant

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

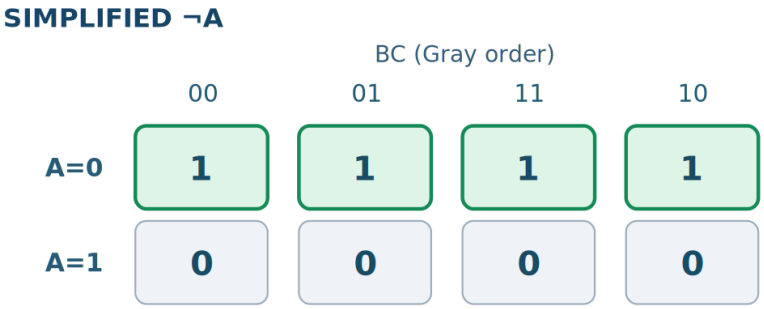
Problem

Four 1s fill the entire A=0 row. Simplify the output.

Follow the worked method

1. Predict and explain — B and C each vary across the row; neither is needed in the expression.
2. Check and extend — A stays 0 in the whole group. The simplified term is ¬A.

Source model · final illustrated step



Group adjacent 1s; retain only variables that stay constant

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict and explain
A changes between rows, and C changes between columns 11 and 10.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Why do we group neighbouring 1s in a Karnaugh map?

Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A group of four 1s can remove two changing variables.

Use key vocabulary from memory

1. Explain Karnaugh map in your own words.

2. Explain Gray code in your own words.

3. Explain group in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.